How many copies is coherence worth?

Exact exchange rates between two-copy and single-copy readout of quantum data

[Author list to be completed]

Abstract. Coherent access to two copies of a quantum state estimates nonlinear properties with far fewer measurements than single-copy readout, but how many fewer *state preparations* it saves is usually stated asymptotically or as an upper bound, not at the budget of an experiment. We measure and predict this exchange rate for the purity $\mathrm{tr}(\rho_A^2)$ of a subsystem, the building block of Rényi entropies and of concentratable entanglement. At $C = 400$ copies per state and $|A| = 4$, two-copy Bell sampling reaches an RMSE of 0.051 against 0.194 for the strongest local single-copy reader, in all ten pre-registered seeds (exact Wilcoxon $p = 9.8 \times 10^{-4}$, Holm-adjusted 0.002). To match the coherent reader, local randomized measurements need 7.8 times more copies at $|A| = 4$ and 191 at $|A| = 8$; global random measurements, which scale better with the dimension, still need 19.3. Evaluating the known U-statistic variance decomposition exactly, instead of bounding it, predicts these rates without fitted parameters and shows that the published bound overstates them four- to fivefold at our budget and up to seventeenfold asymptotically (704 against 42 at $|A| = 8$). The rate depends on the budget: it is largest where experiments operate and grows as $\approx 2.7^{|A|}/\sqrt{C}$ there. The advantage persists when the label must be learned (circuit depth, von Neumann entropy), on the NTangled dataset, under exact depolarising and readout noise, where both readers share the bias but only the coherent one calibrates itself for free, and on an emulated IBM Heron device with calibrated noise.

Keywords: quantum data, randomized measurements, classical shadows, Bell sampling, sample complexity, purity, entanglement.

Introduction

Machine learning on quantum data has a currency problem. Two readers of a quantum state can be compared at equal shots, equal parameters or equal circuit depth, and the choice decides the outcome. Separations that survive scrutiny are stated in *copies*, the number of times the unknown state is prepared, because that is what a single-copy device and a coherent multi-copy device both consume[1,2]. Classical shadows predict many properties from few single-copy measurements[3], classical learners on shadows are provably efficient for broad classes of many-body problems[4], and quantum convolutional networks on standard phase datasets are effectively simulable from Pauli shadows[5]. A single-copy baseline is therefore not optional.

Nonlinear functionals are where the currency bites. Randomized measurements estimate Rényi-2 entropies from correlations between single-copy outcomes[6–8]; the variance of the resulting U-statistic splits into a term linear and a term quadratic in the snapshots, with published upper

bounds exponential in the subsystem size[9]. The optimal single-copy sample complexity of purity estimation scales as $\sqrt{d}$[10,11], whereas the swap observable measured transversally on two copies estimates it at a cost independent of the dimension (Methods). One extra replica gives exponential savings for nonlinear properties[12], an infinite hierarchy of multi-copy tasks exists[13], conjugate states and minimal quantum memory recover part of the gain[14], and purification with constant ancillas gives exponential separations[15]. In the other direction, quantum PAC learning gains require inverse access to the preparation[2] and bound entanglement does not suffice for an exponential learning advantage[16]. Experiments have demonstrated multi-copy neural networks on two qubits[17] and entanglement-enhanced learning on up to sixteen qubits[18]; two-copy shadow tomography has been evaluated for estimation rather than learning[19], and learning with optimal similarity measurements[20] and phase classification from partial tomography[21] have been proposed.

These results are asymptotic statements, upper bounds, or comparisons at unequal budgets. A practitioner who must decide whether to spend a qubit register on a second copy needs a number: at the budget available, for the subsystem of interest, how many single-copy preparations does coherence replace? We provide that number for local and global single-copy readers, show how to predict it exactly from the state ensemble, and test whether it survives learning, a published dataset, noise and compilation to a current device.

Contributions. (i) A pre-registered equal-copy comparison against the strongest single-copy readers we could build, local and global. (ii) A measured exchange rate for $|A| = 2$ to 8, together with an exact evaluation of the U-statistic variance for the actual ensemble which predicts it without fitted parameters, shows that the published bounds overstate it severalfold, and makes explicit that it depends on the budget. (iii) Evidence that the advantage survives when a learner must combine features into a label no reader estimates directly, and on NTangled at equal copies. (iv) An exact noise study and a transpiled, calibrated emulation on an IBM Heron model, with a copy and circuit budget for a hardware run. (v) An audited protocol whose corrections are reported.

Research questions

RQ1. At equal copies, how much more accurately does a two-copy reader estimate subsystem purity than the strongest local single-copy reader?
RQ2. How many single-copy preparations does coherence replace, as a function of $|A|$, of the budget and of the single-copy reader, and can the number be predicted?
RQ3. Does the advantage survive when the label must be learned, and on a published dataset?
RQ4. Does it survive realistic noise and compilation to a current device?

Results

Readers and budget

All readers receive the same C copies of each state. The *coherent reader* measures $C/2$ pairs transversally in the Bell basis; the product of swap signs over a window W is a ± 1 variable whose

mean is $\mathrm{tr}(\rho_W^2)$, and one sample set yields every window purity at once (Methods). The *local single-copy reader* measures each copy in a random local Pauli basis and estimates the purity with the unbiased U-statistic over pairs of snapshots[3,8]. One shot per basis, clipped to the physical range $[2^{-|A|}, 1]$, is its most copy-efficient configuration in our tests (Supplementary Table S1); the grouped configuration with ten shots per basis, common on hardware, is reported alongside. The *global single-copy reader* applies a Haar-random unitary to all of A before each one-shot measurement; it has the better dimension scaling but requires an entangling circuit across A, whereas both the local and the Bell reader use depth-one measurement circuits. States are brickwork circuits of depth one to four with Haar two-qubit gates on $n = 2|A|$ qubits, and A is the left half.

Confirmatory comparison

Ten seeds fixed in advance, 400 states each, at $|A| = 4$ and $C = 400$ (Table 1). The coherent reader reaches an RMSE of 0.049–0.053; the strongest local single-copy reader reaches 0.180–0.201. The paired difference is -0.143 (s.d. 0.006), negative in all ten seeds; the exact one-sided Wilcoxon test gives $p = 9.8 \times 10^{-4}$, the smallest value attainable with ten seeds, and Holm correction over the pre-registered two-idea family gives 0.002. Against the grouped reader the factor is 6.6 instead of 3.8. The campaign passes in all ten seeds on PennyLane and on Qiskit, whose Bell probabilities agree to 2.2×10^{-16}.

Table 1: Confirmatory campaign: RMSE of $\mathrm{tr}(\rho_A^2)$ at $C = 400$, $|A| = 4$, $n = 8$, 400 states per seed. "Strongest" is the minimum over the local single-copy readers (in every seed, one shot per basis, clipped); "grouped" is the pre-registered ten-shot reader.

Seed	Two-copy Bell	Strongest single-copy	Grouped ($S = 10$)	Difference
0	0.0510	0.1946	0.3861	−0.1437
1	0.0501	0.2011	0.3365	−0.1509
7	0.0521	0.1931	0.3546	−0.1411
42	0.0505	0.1942	0.3127	−0.1437
123	0.0530	0.1924	0.3439	−0.1394
456	0.0526	0.1966	0.3621	−0.1440
789	0.0491	0.2007	0.3208	−0.1516
1234	0.0498	0.1797	0.3198	−0.1299
2024	0.0509	0.1943	0.3249	−0.1434
3407	0.0518	0.1907	0.3339	−0.1389
Mean	0.0511	0.1937	0.3395	−0.1426

The exchange rate

Figure 1a sweeps $|A| = 2, \ldots, 8$ at $C = 400$ (n up to 16, ten seeds). The coherent reader stays at 0.048–0.052. The local reader degrades from 0.078 to saturation near the physical bound and is worse than a constant predictor from $|A| = 5$. The global reader degrades more slowly, from 0.071 to 0.44. Converting these errors into copies requires care, because the single-copy curves are not parallel to the coherent one (panel b): Over the measured range the single-copy RMSE falls as $C^{-0.60}$ between $C = 100$ and 800 and as $C^{-0.65}$ from $C = 1600$ until it reaches the coherent reader,

faster than $C^{-1/2}$, because clipping to the physical range removes a share of the error that shrinks as estimates move back inside it; the coherent reader follows $C^{-0.52}$ throughout. We therefore measure the copies each reader needs on a grid of budgets and interpolate (panel c). Matching the coherent reader at $C = 400$ takes the local reader 2.6, 4.8, 7.8, 16.6, 32, 76 and 191 times more copies for $|A| = 2, \ldots, 8$, and the global reader 2.0, 2.7, 3.8, 4.9, 7.0, 10.6 and 19.3 times more. Coherence therefore beats every non-adaptive single-copy reader we tested at every size, by a factor that grows with the subsystem; the global reader's slower growth is consistent with the $\sqrt{d}$ scaling of optimal single-copy purity estimation [10,11].

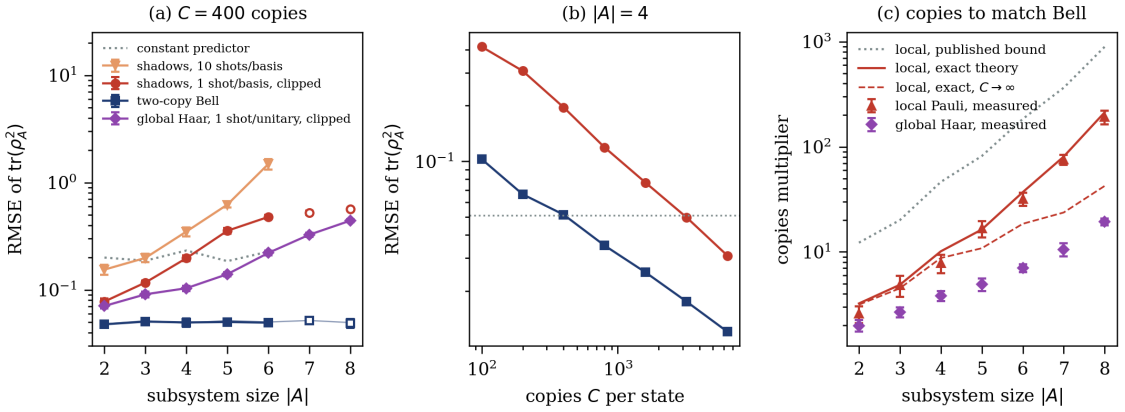


Figure 1: The exchange rate. (a) RMSE of $\mathrm{tr}(\rho_A^2)$ against subsystem size at $C = 400$; mean and s.d. over ten seeds; open markers at $|A| = 7, 8$ come from the copies sweep on the same states; thin line: closed-form coherent prediction; dotted: constant predictor. (b) RMSE against copies at $|A| = 4$; the dotted line marks the coherent reader at $C = 400$. (c) Copies a single-copy reader needs to reach the coherent reader's RMSE at $C = 400$: measured on a grid of budgets and interpolated (markers; mean and s.d. over ten seeds), exact theory for the local reader at $C = 400$ (solid) and $C \to \infty$ (dashed), Eq. (2), and the rate implied by the published variance bound [9] (dotted).

Exact prediction. The coherent estimator is the mean of $C/2$ independent signs, so $\mathrm{Var_B} = 2(1-p^2)/C$ with $p = \mathrm{tr}(\rho_A^2)$, independent of $|A|$. The local estimator is a second-order U-statistic over M snapshots with kernel $h(x,y) = \mathrm{tr}(\hat{\sigma}_x \hat{\sigma}_y)$, and its variance has the standard decomposition [9]

$$\mathrm{Var_S}(M) = \frac{4(M-2)}{M(M-1)}\,\zeta_1 + \frac{2}{M(M-1)}\,\zeta_2, \qquad \zeta_1 = \mathrm{Var}_x\,\mathrm{tr}(\hat{\sigma}_x \rho_A), \quad \zeta_2 = \mathrm{Var}_{x,y}\,h(x,y). \tag{1}$$

Ref. 9 bounds the two terms by $\zeta_1 \le 2^{|A|} p$ and $\zeta_2 \le 2 \cdot 8^{|A|}$. We instead evaluate them exactly for every state, as contractions of the $6^{|A|}$ table of snapshot probabilities (Methods). Equating dataset-averaged variances gives the exchange rate $m = M/C$,

$$m \simeq \begin{cases} \sqrt{\overline{\zeta_2} \,/\, \left(\overline{1-p^2}\, C\right)} & M \ll M^* = \zeta_2 / 2\zeta_1, \\ 2\,\overline{\zeta_1} \,/\, \overline{1-p^2} & M \gg M^*, \end{cases} \tag{2}$$

solved exactly by root finding in practice. Without fitted parameters, the predicted RMSE at $C = 400$ matches the simulated one within 4% for both readers at every $|A| = 2, \ldots, 6$, and the

predicted rates (3.2, 4.9, 10.1, 16.3, 37, 81, 213) agree with the measured ones within 30%, lying above them in six of seven sizes because clipping only helps the measured reader.

Two consequences follow. First, the published bound is loose for this ensemble: it implies 46 instead of 10.1 at $|A| = 4$ and 895 instead of 213 at $|A| = 8$, and 704 instead of 42 asymptotically. Over our states ζ_2 grows by 7.2 per qubit, close to the second moment of the single-qubit kernel, which equals 7 for a maximally mixed qubit, against 8 in the bound; ζ_1 grows by only 1.4–1.8, against 2. Second, the rate depends on the budget. The crossover M^* rises from 35 snapshots at $|A| = 2$ to 3.4×10^5 at $|A| = 8$, so beyond $|A| \approx 4$ every practical budget sits in the first regime, where the rate grows as $\approx 2.7^{|A|}$ and falls as $C^{-1/2}$. The price of coherence is largest exactly where experiments operate, and it can be computed from a model of the state ensemble before any measurement is taken.

Labels that must be learned, and a published dataset

A threshold on an estimated purity is estimation with a decision rule attached. We therefore pre-registered two tasks in which a learner must combine features into a target that no reader estimates directly (Methods): (L1) recovering the preparation depth, four classes, from all contiguous window purities; (L2) regressing the von Neumann entropy $S(\rho_A)$, which is not a polynomial in the state. Every reader spends $C = 400$ copies per state and feeds the same learner; we add the shadow-ML baseline of local Pauli expectations[4].

Table 2: Learned labels, $n = 8$, $C = 400$, ten official seeds, 160 test states each; mean (s.d.). L1: four-class depth error (chance 0.75). L2: RMSE of $S(\rho_A)$ in bits, $|A| = 4$ (target spread 0.61). "Exact" uses noiseless window purities and is a ceiling, not a reader.

Features	L1 depth error	L2 entropy RMSE
Exact window purities (ceiling)	0.024 (0.013)	0.052 (0.003)
Two-copy Bell, window purities	**0.049** (0.026)	**0.148** (0.011)
Shadows, 1 shot/basis, clipped	0.112 (0.028)	0.215 (0.020)
Shadows, 10 shots/basis	0.139 (0.048)	0.247 (0.012)
Shadows, local Pauli $k \leq 2$	0.753 (0.032)	0.833 (0.048)

The coherent reader halves the depth error of the best single-copy reader (-0.062 ± 0.027) and lowers the entropy RMSE by a third (-0.067 ± 0.019), in all ten seeds for both tasks (exact Wilcoxon $p = 9.8 \times 10^{-4}$ each; exploratory family, Table 2). Local Pauli expectations are at chance on both: a linear model on them cannot represent functionals quadratic or higher in the state. On L2 the coherent reader remains far from the ceiling (0.148 against 0.052), because regressing an entropy from purities amplifies estimation noise.

NTangled[22] labels four-qubit states by their concentratable entanglement $\mathrm{CE} = 1 - 2^{-n} \sum_{S \subseteq [n]} \mathrm{tr}(\rho_S^2)$[23], a sum of subset purities that one Bell sample set delivers in full. Rebuilding the released generators (Methods), at $C = 400$ copies, ten seeds and 200 states per seed, the coherent reader estimates CE with an RMSE of 0.024 ± 0.002 against 0.054 ± 0.003 for the strongest single-copy reader, and classifies the two classes with an error of 0.038 ± 0.030 against 0.114 ± 0.052, lower in all ten

seeds on both metrics ($p = 9.8 \times 10^{-4}$), with the exact CE reaching 0.029 (Fig. 2). The original NTangled two-copy experiment doubled the copy budget[22]; at equal copies the gain persists.

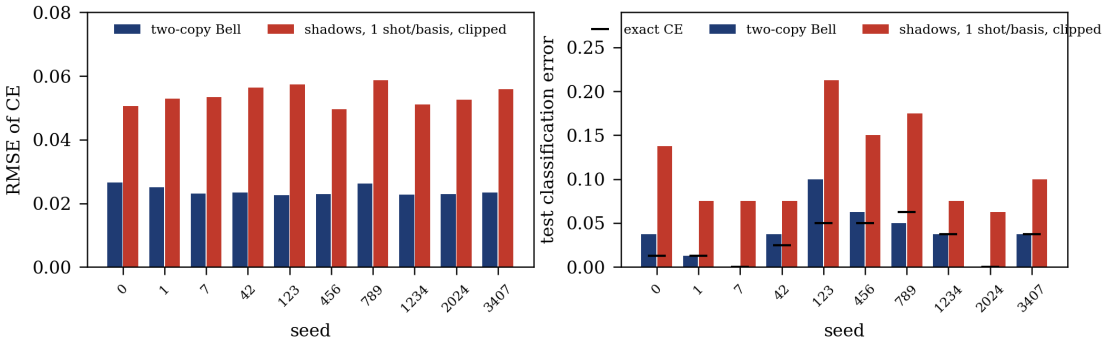


Figure 2: NTangled, four qubits, classes with target CE 0.05 and 0.25, $C = 400$ copies, per seed. Left: RMSE of the concentratable entanglement. Right: test error of a threshold classifier on the estimate; black ticks mark the exact CE.

Noise and an emulated device

We apply per-qubit depolarising noise p to every qubit of every copy and readout error ε to every measured bit, simulated exactly on the raw outcomes (Methods; Fig. 3, Supplementary Table S2). Depolarising noise biases both readers by the same amount (-0.248 and -0.244 at $p = 0.10$), as expected from unbiased estimators of the purity of the noisy state, so noise erodes the coherent reader's relative lead without singling it out. Uncorrected, it stays ahead in all ten seeds up to $p = 0.20$ and $\varepsilon = 0.20$, and in three of ten at $p = 0.30$. What separates the readers is calibration: for pure input states the purity of the whole register is one, so the coherent reader reads its contraction off the same samples at no cost and inverts it (Methods), whereas the single-copy reader would need the purity of all $2|A|$ qubits. Self-calibrated, the coherent reader stays ahead in all ten seeds over the whole grid, by $3.1\times$ at $p = 0.10$ and $3.3\times$ at $\varepsilon = 0.10$. Memory dephasing has no effect: purity is invariant under local unitaries and local Paulis map Bell states to Bell states.

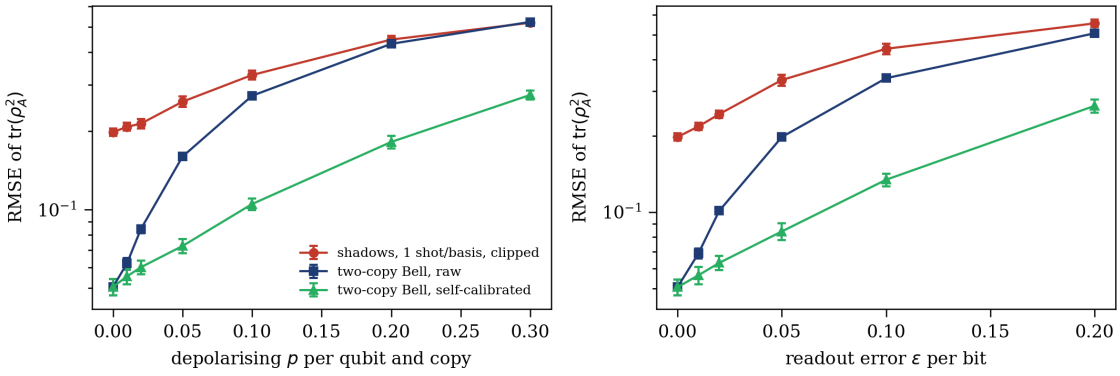


Figure 3: Exact noise at $|A| = 4$, $C = 400$; mean and s.d. over ten seeds. Left: per-qubit depolarising error on both copies. Right: readout error per measured bit. Noise is simulated on the raw outcomes and validated against density-matrix calculations (Methods).

To include compilation and a realistic error budget, we transpiled both protocols to the heavy-hex `FakeFez` model (156-qubit IBM Heron r2) at optimisation level 3 and simulated them with its calibrated noise model: gate and readout errors and relaxation and dephasing times, plus the routing overhead of the connectivity (Methods). The two-copy reader runs on $2n$ physical qubits with $C/2$ shots; the local single-copy reader runs one random Pauli basis per shot.

Table 3: Transpiled emulation on the IBM Heron r2 model `FakeFez`, $C = 400$ copies per state, $n = 2|A|$; RMSE of $\mathrm{tr}(\rho_A^2)$, mean (s.d.) over official seeds. 2q gates: mean two-qubit gate count after transpilation (Bell / one single-copy circuit). Wins: seeds in which the self-calibrated Bell reader beats the single-copy reader.

$\lvert A\rvert$	Noise	Seeds	Bell raw	Bell calibrated	Single-copy	2q gates	Wins
2	FakeFez noise	10	0.077 (0.006)	0.060 (0.007)	0.089 (0.006)	41 / 12	10/10
2	ideal	10	0.049 (0.005)	0.049 (0.005)	0.077 (0.007)	41 / 12	10/10
3	FakeFez noise	10	0.108 (0.007)	0.077 (0.006)	0.138 (0.013)	69 / 19	10/10

At $|A| = 2$ the self-calibrated coherent reader reaches an RMSE of 0.060 against 0.089 for the single-copy reader (factor 1.5, lower in 10 of 10 seeds, exact Wilcoxon $p = 0.00098$); uncorrected it reaches 0.077, and the calibration constant is $\lambda = 0.949$ per qubit. At $|A| = 3$ the self-calibrated coherent reader reaches an RMSE of 0.077 against 0.138 for the single-copy reader (factor 1.8, lower in 10 of 10 seeds, exact Wilcoxon $p = 0.00098$); uncorrected it reaches 0.108, and the calibration constant is $\lambda = 0.945$ per qubit. The Bell circuit is several times deeper in two-qubit gates than a single-copy circuit, because it prepares two copies and routes the pairwise CNOTs on heavy-hex; its estimate nevertheless stays ahead in every seed once the shared contraction is calibrated out.

The configuration at $|A| = 2$ is the 2×4-qubit proof of concept we propose for hardware. With 100 states it needs 100 Bell circuits of 200 shots and 900 single-copy circuits sharing 40,000 shots: 6×10^4 shots and 10^3 circuits in total, a single short job on current IBM devices.

Discussion

Coherent two-copy readout is worth a measurable, predictable number of preparations for labels built from subsystem purities. Against local randomized measurements the number is about three at $|A| = 2$ and about two hundred at $|A| = 8$ for our budget; against global random measurements it is two and about twenty. The comparison is fair in the sense that matters: the budget is in copies, the single-copy readers are the most copy-efficient configurations we found, local and global, the learner on top is the same, and the noise is simulated exactly.

The exchange rate is a property of the state ensemble and the budget, not a constant. It follows from the U-statistic variance decomposition[9] once the two terms are evaluated for the states at hand rather than bounded, and the bound overstates it four- to fivefold at our budget and by more than an order of magnitude asymptotically. A practitioner can thus compute the rate for a target ensemble before deciding to spend qubits on a second copy. Against global single-copy readers the rate is

smaller but still exceeds one at every size, before counting the entangling circuits across A that those readers need.

The gap is statistical rather than representational. A classical model on all 3^4 full-support Pauli expectations represents the purity exactly and still fails at this budget (Supplementary Table S1), and the learned tasks show that the gap is not an artefact of thresholding a single estimate. What our results do not establish is an advantage for labels outside the span of low-order purities, such as full spectra, where a trainable multi-copy circuit would be the object of study, nor for labels linear in the state, where single-copy readers are already optimal. Adaptive single-copy protocols can reach the optimal $\sqrt{d}$ scaling[11]; our global reader is non-adaptive, so the exchange rate against the best adaptive single-copy protocol may be lower than the global rates we report.

Three limitations bound the claims. The self-calibration relies on pure input states; for mixed-state data the anchor disappears. No physical device was used: the noise thresholds and the emulation are design targets, and the emulated model omits crosstalk, leakage and drift. And the analyses beyond the confirmatory campaign (Fig. 1, Tables 2 and 3) were added after it; they use the official seeds, are labelled exploratory and all favour the coherent reader in every seed, but they are not covered by the pre-registered test.

Our process should also be stated. The confirmatory endpoint moved from classification error to estimator RMSE after the single-seed gate and before any confirmatory seed, because development seeds showed that an 80-state test split could not resolve the pre-registered margin. An internal audit after the campaign found that the pre-registered single-copy reader was not the strongest available and that an earlier noise model, acting on swap signs instead of measured bits, was not exact and was biased against the coherent reader. Both are corrected here; the campaign is re-scored against the stronger reader, which only makes the test harder. The first 200 of the 400 states of seed 42 coincide with the gate's states.

Methods

States, seeds and statistics

Brickwork circuits of depth $d \in \{1,2,3,4\}$, drawn uniformly, with Haar-random two-qubit gates on a chain of $n = 2|A|$ qubits; A is the left half. All results use the ten seeds 0, 1, 7, 42, 123, 456, 789, 1234, 2024 and 3407, fixed before any confirmatory run; development used seeds ≥ 9001 (Supplementary Table S1). Per-seed paired differences (coherent minus strongest single-copy), exact one-sided Wilcoxon signed-rank test, paired t test for sensitivity, Holm correction over the pre-registered two-idea family, $\alpha = 0.05$; ten seeds make 9.8×10^{-4} the smallest attainable p.

Two-copy reader

Each shot measures qubit q of both copies in the Bell basis (CNOT from copy 1 to copy 2, Hadamard on copy 1, readout of bits a_q, b_q). The sign $s_q = (-1)^{a_q b_q}$ is the eigenvalue of the swap on pair q and $\mathrm{tr}(\rho_W^2) = \mathbb{E}[\prod_{q \in W} s_q]$ for every subset W. For noise-free sweeps we use the exact equivalent that the

product over A is a ± 1 variable with mean p, i.e. $2\,\mathrm{Bin}(C/2,(1+p)/2)/(C/2)-1$; the campaign, noise, learning, NTangled and emulation experiments sample the full $2n$-qubit Bell distribution.

Single-copy readers

Local. C snapshots in C/S random local Pauli bases with S shots each; unbiased U-statistic over pairs from different bases (same-basis pairs are correlated and excluded). For noise-free sweeps snapshots restricted to A are drawn from ρ_A directly, which is exact for local measurements, using the exact outcome table of all $3^{|A|}$ bases computed by contracting ρ_A qubit by qubit; an oracle checks the table against explicit unitaries and the fast estimator against the reference implementation (to 10^{-9}) before every run. *Global.* One Haar-random unitary on A per shot, snapshot $(d+1)U^\dagger|b\rangle\langle b|U-I$, so $\mathrm{tr}(\hat\rho_i\hat\rho_j)=(d+1)^2|\langle v_i|v_j\rangle|^2-(d+2)$ with $v=U^\dagger|b\rangle$ and the U-statistic follows from $\|\sum_i v_i v_i^\dagger\|_F^2$. The observed v has density $d\langle v|\rho_A|v\rangle$ on the sphere, a mixture over eigenvectors e_j with weights λ_j of densities $d|\langle v|e_j\rangle|^2$, under which $|\langle v|e_j\rangle|^2\sim\mathrm{Beta}(2,d-1)$; we sample it exactly in $O(d)$ per shot and check it against explicit Haar unitaries and a rejection sampler. All estimates are clipped to $[2^{-|A|},1]$.

Exchange rates

Copies to match the coherent reader at $C=400$ are measured on a grid of budgets (C doubling from 100) and interpolated in log–log, never extrapolated. For the exact prediction, the snapshot distribution $P(x)$ over $x=(b_q,o_q)_{q\in A}$ ($6^{|A|}$ symbols) is computed by contracting ρ_A with the basis rotations. With the single-qubit kernel T (5 for equal basis and outcome, -4 for equal basis and different outcome, $1/2$ otherwise), $g=T^{\otimes|A|}P$, $p=P^\top g$, $\zeta_1=P^\top g^2-p^2$ and $\zeta_2=P^\top(T^{\circ 2})^{\otimes|A|}P-p^2$. The formula Eq. (1) is checked against 4000 Monte Carlo repetitions before every run (within 8%). The bound-implied rate replaces $4\zeta_1/M+2\zeta_2/M^2$ by $4\cdot 2^{|A|}p/M+4\cdot 8^{|A|}/M^2$ (Eq. D16 of Ref. 9).

Noise and self-calibration

Depolarising error p on a qubit is a uniformly random Pauli, each with probability $p/4$; through the Bell circuit X on either copy flips b_q, Z flips a_q, Y flips both. Readout flips each bit with probability ε. For the local reader a depolarised qubit gives a uniform outcome and readout flips the bit. Both models act on raw outcomes of clean samples, which is exact for these channels, and agree with density-matrix calculations on $n=2$ to within 7×10^{-4}. We model the contraction as $\mathbb{E}[\hat p_W]\approx\lambda^{|W|}p_W+(1-\lambda^{|W|})2^{-|W|}$, estimate one global λ from the full-register Bell purity, $\lambda=\big((\overline{\hat p}_{\mathrm{full}}-2^{-n})/(1-2^{-n})\big)^{1/n}$, and invert for A. A per-state λ amplifies calibration noise and is not used; without noise the correction is the identity.

Learned labels

Pre-registered in `exp_bell_learn.py` before any run; development seed 9501 only checked that the pipeline executes. $n=8$, 400 states per seed, 240/160 split, $C=400$ copies per state for every

reader. Features: the 26 contiguous window purities of width 1 to 4, clipped, and their logarithms; the local-Pauli baseline uses all weight-≤ 2 expectations from the same snapshots. L1: multinomial logistic regression ($C = 1$); L2: ridge regression ($\alpha = 1$); standardised features, no hyperparameter search.

NTangled

Weights from `LSchatzki/NTangled_Datasets`, hardware-efficient ansatz, four qubits, depth three: per layer $U3$ on every qubit, CZ on $(2i, 2i+1)$, $U3$ again with the same angles, CZ on $(2i+1, 2i+2)$, with $U3(\theta,\phi,\lambda) = R_z(\phi)R_x(-\pi/2)R_z(\theta)R_x(\pi/2)R_z(\lambda)$. Inputs are product states with real amplitudes, which reproduce the target CE of each class (exact means 0.049 and 0.252 for the classes labelled 0.05 and 0.25). One hundred states per class, $60/40$ split, logistic regression on the scalar CE estimate.

Device emulation

`FakeFez` from `qiskit-ibm-runtime` and `NoiseModel.from_backend` in Aer; transpilation at level 3 with the seed as transpiler seed. One shot per random basis is implemented exactly by drawing the C basis labels first and running each distinct basis with its multinomial count. Bell and single-copy circuits are simulated in separate batches with an explicit method: a single batch mixing them returned near-constant outcomes for the single-copy circuits, while each circuit alone was correct. A noiseless run of the same pipeline reproduces the ideal results (Table 3).

Data availability

NTangled generator weights are public (https://github.com/LSchatzki/NTangled_Datasets). All other data are generated from the listed seeds; per-seed results are in `results/bell_qnn/`.

Code availability

Repository `qdata-gates`: `exp_bell_qnn.py` (gate and campaign), `exp_bell_paper.py` (scaling, copies, noise, NTangled), `exp_bell_global.py` (global reader and bound), `theory_bell.py` (exact variances), `exp_bell_learn.py` (learned labels), `exp_bell_emulated.py` (device emulation), `qdlib.py` (estimators), `audit_*.py`, `fig_paper.py` and `fill_macros.py`, which writes every swept number in this manuscript from the result files.

Author contributions

To be completed.

Competing interests

The authors declare no competing interests.

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Supplementary information

Table S1: Development comparison of single-copy readers: classification error at $C = 400$ copies, mean over ten development seeds, 80 test states each, local readers with $S = 10$. The coherent reader is a logistic regression on all Bell window purities. A classical model on all 3^4 full-support Pauli expectations represents the purity exactly, $\mathrm{tr}(\rho_A^2) = 2^{-|A|}\sum_P \langle P\rangle^2$, and still fails at this budget. Grouping also costs accuracy: the purity RMSE at $|A| = 4$ was 0.332 ($S = 10$), 0.274 ($S = 5$), 0.245 ($S = 2$) and 0.234 ($S = 1$), and clipping lowered the last to 0.202.

Reader	Test error
Two-copy Bell, window purities	**0.086**
Shadows, window purities (U-statistics)	0.135
Shadows, local Pauli $k \leq 2$, logistic	0.494
Shadows, full support on A (3^4 chains), logistic	0.519
Shadows, full support on A, SVM-RBF	0.403
Inverted shadow density matrix, $\mathrm{tr}(\hat{\rho}_A^2)$	0.256
Hybrid: U-statistic purity + full-support chains	0.404

Table S2: Noise study, $|A| = 4$, $C = 400$, mean over ten seeds. "Wins" counts seeds in which the coherent reader (raw / self-calibrated) has lower RMSE than the strongest local single-copy reader.

Channel	Level	Bell	Bell cal.	Single	Bias Bell	Bias single	Wins
none	0	0.051	0.051	0.198	+0.001	−0.038	10 / 10
depolarising	0.02	0.084	0.060	0.213	−0.059	−0.084	10 / 10
depolarising	0.05	0.160	0.073	0.259	−0.139	−0.149	10 / 10
depolarising	0.10	0.273	0.105	0.328	−0.248	−0.244	10 / 10
depolarising	0.20	0.431	0.182	0.448	−0.398	−0.381	10 / 10
depolarising	0.30	0.523	0.275	0.519	−0.485	−0.456	3 / 10
readout	0.02	0.101	0.063	0.244	−0.078	−0.128	10 / 10
readout	0.05	0.198	0.084	0.332	−0.175	−0.249	10 / 10
readout	0.10	0.338	0.134	0.441	−0.309	−0.376	10 / 10
readout	0.20	0.508	0.263	0.556	−0.470	−0.497	10 / 10